

Brief summary of this function: Plots all of the subplots in the main live script at once for the different phases of Energy vs Time.

Detailed explanation of this function: This live script helper function was made to make the readability of the main script easier, so there isn't numerous lines for plotting (ie: subplot, plot, title) in the main code. It is made to plot Energy vs Time in the different phases. This function starts off by defining the modes in questions, than creating plots for them in one figure. It starts off with CC, CV, Rest, then Undefined phases.

```
function plotEnergy_phases(time, energy, modes) %naming the function

%defining the different phases
CC = modes == "CC";
CV = modes == "CV";
Rest = modes == "Rest";
Undefined = modes == "Undefined";

%creating the plots
figure

% plotting the CC graph for energy vs time
%using basic plotting techniques used in other helper functions/main script
%ie: subplot, plot, title, labeling axes
subplot(2,2,1)
plot(time(CC), energy(CC), 'Color', [0.5,0,0]) %choosing the same red color
for consistency

title('Energy vs Time (CC)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on

%plotting the CV graph
subplot(2,2,2)
plot(time(CV), energy(CV), 'b') %choosing the same blue color for consistency

title('Energy vs Time (CV)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on

%plotting the graph during the Rest phase
subplot(2,2,3)
plot(time(Rest), energy(Rest), 'g') %choosing same color for rest (green)

title('Energy vs Time (Rest)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on

%plotting graph during the Undefined regions
```

```
subplot(2,2,4)
plot(time(Undefined), energy(Undefined), 'c') %choosing same color for
undefined (cyan)

title('Energy vs Time (Undefined)')
xlabel('time [s]')
ylabel('Energy [W]')
grid on

sgtitle('Energy vs Time Graphs for the Seperate Phases') %main title for the
subplots

end
```